Election-night model: specification

Young-Kyung Kim

October 20, 2026

1 Outcomes

The model estimates three outcomes at the precinct level, separately by vote mode, giving nine equations in total. Each equation is estimated separately, but the errors are modeled jointly through a 9×9 covariance matrix Σ.

Let $m \in \{\text{mail}, \text{early}, \text{eday}\}$, and let v_{im}^{dem}, v_{im}^{rep}, v_{im}^{oth} denote votes cast in mode m in precinct i for the Democratic candidate, the Republican candidate, and all others.

- Three equations for τ_{im}, total votes cast (turnout) in the cell;
- Three equations for tp_{im}, the third-party share of the cell;
- Three equations for s_{im}, the two-party Democratic share of the cell.

What Σ does. Each equation is estimated separately; since they share a design matrix, seemingly unrelated regression coincides with equation-by-equation estimation, so Σ does not change $\hat{\beta}$. It should improve prediction; the aggregate is jointly nonlinear in (τ, tp, s), so the predictive variance depends on their correlation. With the plausibility of correlation between vote shares and turnout by vote modes in precincts, this should mean the conditional covariance is smaller than the unconditional one. Knowing that we might want individual precinct predictions also motivated this modeling choice: I'd prefer to have more information to pull into those more weakly powered estimates.

2 The factorization

Predicting v^{dem}, v^{rep} and v^{oth} separately and summing them can produce impossible values — shares that do not sum to one, negative counts. Clamping such values afterwards would spend researcher degrees of freedom we would rather not spend. Instead we change variables. The map

$$\tau_{im} = v_{im}^{\text{dem}} + v_{im}^{\text{rep}} + v_{im}^{\text{oth}}, \qquad tp_{im} = \frac{v_{im}^{\text{oth}}}{\tau_{im}}, \qquad s_{im} = \frac{v_{im}^{\text{dem}}}{v_{im}^{\text{dem}} + v_{im}^{\text{rep}}}$$

is a bijection, so the quantity we care about can be written as a function of (τ, tp, s) and recovered exactly. Every draw is a valid election outcome by construction, with no clamping and no renormalizing.

3 Link functions

The two share outcomes are bounded in $[0, 1]$, so both are modeled on the logit scale:

$$y^{tp}_{im} = \text{logit}\left(\frac{v^{\text{oth}}_{im} + \frac{1}{2}}{\tau_{im} + 1}\right), \qquad y^{s}_{im} = \text{logit}\left(\frac{v^{\text{dem}}_{im} + \frac{1}{2}}{v^{\text{dem}}_{im} + v^{\text{rep}}_{im} + 1}\right).$$

The $\frac{1}{2}$ offsets keep the transform finite where a third-party candidate receives no votes in a precinct, which is common.

Turnout is modeled on a log scale, because precinct-level mode-specific turnout is right-skewed. Where the composition is disabled the outcome is simply $y^{\tau}_{im} = \log(\tau_{im} + 0.1)$, the offset keeping cells with zero counted votes finite. Getting this scale right mattered more than expected.

4 Turnout: total and composition

By default turnout is reparameterized rather than modeled as three free log-levels, because turnout has a known ceiling and the three modes must add up. Writing T_i for the precinct total and p_{im} for its split across modes,

$$\theta_i = \text{logit}\left(\frac{T_i}{\kappa n_i}\right), \qquad \eta_{im} = \log\frac{p_{im}}{p_{i,\text{ref}}} \quad (m \neq \text{ref}), \qquad p_i = \text{softmax}(\eta_i), \qquad \tau_{im} = T_i\, p_{im}, \tag{1}$$

with $\kappa = 1.25$ and n_i the precinct's registrants. The fitted outcomes are θ_i in the reference mode's slot and the η_{im} in the other two, so the equation count is unchanged at $1 + (M - 1) + 2M = 3M$.

This buys two guarantees. First, $\sum_m p_{im} = 1$ gives $\sum_m \tau_{im} = T_i$ exactly, so a surplus in one mode can no longer inflate the precinct total. Second, $0 < T_i < \kappa n_i$. An earlier version of the model predicted turnout at 184% of truth; under (1) that is arithmetically impossible rather than something detected after the fact.

Setting ref = early is an identifiability normalization; early is chosen because it reports first and is the best-measured mode.

5 Partial counts

Treating an incomplete snapshot as final, or thresholding it as "complete enough," produces strange behavior, and the problem compounds when the decision is made per precinct-mode rather than per precinct. We therefore model partial counts as a mixture. With c_{im} the ballots already counted in a cell and r_{im} those remaining,

$$r_{im} = \tau_{im} - c_{im}, \qquad \text{dem}_{im} = c^{\text{dem}}_{im} + r_{im}\,(1 - tp_{im})\,s_{im}. \tag{2}$$

A counted-out cell has $r_{im} \approx 0$, so its observed share is pinned automatically; a barely-started cell is mostly predicted. No completeness classifier is required.

Partial counts also impose their own restrictions, and these are inequalities. The floor must act on the precinct *total*: flooring each cell at its own count destroys the adding-up property the parameterization exists for. So we floor the total and hand the remainder only to modes short of their composition-implied share, in proportion to how short they are:

$$\begin{aligned} T_i &\leftarrow \max\left(T_i,\ \textstyle\sum_m c_{im}\right), & \text{short}_{im} &= \max\left(T_i\, p_{im} - c_{im},\, 0\right), \\ r_{im} &= \left(T_i - \textstyle\sum_{m'} c_{im'}\right) \frac{\text{short}_{im}}{\sum_{m'} \text{short}_{im'}}. \end{aligned} \tag{3}$$

This preserves both $\sum_m \tau_{im} = T_i$ and $\tau_{im} \geq c_{im}$.

Known limitation. $\tau_{im} = T_i\, p_{im}$ holds only when no mode has over-reported relative to its composition share, and $T_i < \kappa n_i$ only when $\sum_m c_{im} < \kappa n_i$. Both can fail empirically; the unconditional invariants are $\sum_m \tau_{im} = T_i$ and $\tau_{im} \geq c_{im}$.

6 Covariates and the county effect

- **Turnout and share:** $\{\log n,\ \texttt{turnout22},\ \texttt{turnout24},\ p_{\text{male}},\ p_{65+},\ p_{\text{black}},\ p_{\text{hisp}},\ p_{\text{col}},\ \texttt{dem2}\}$
- **Third party:** $\{\log n,\ p_{\text{male}},\ p_{65+},\ \texttt{dem2},\ \texttt{tp_prior_raw}\}$ — kept deliberately small, since third-party share is a weak and stable signal that overfits easily.

We also include a county random effect to account for dependence across precincts. Ignoring it is not a minor misspecification: measured across nine states, county intraclass correlations run from 0.09 to 0.43, which corresponds to standard errors understated by a factor of roughly 2.4 to 5.1.

7 Estimation

Each equation takes the form

$$y^k_{im} = x_i'\beta^k_m + u^k_{m,c(i)} + \epsilon^k_{im}, \qquad u^k_{m,c} \sim N\big(0,\ \sigma_k^2/\lambda_k\big), \qquad \lambda_k = \frac{1 - \text{ICC}_k}{\text{ICC}_k}. \tag{4}$$

Covariates from the voter file and polling priors supply a fitted prior value for every precinct and mode. We then follow the Bayesian data-augmentation approach used in the NYC mayoral model, with the county random effect added.

Write X for the covariate block and C for the county indicator block. The design stacks N g-prior rows $(X, \mathbf{0})$ at weight n_k/N, the n_{obs} reported rows $(X_{\text{obs}}, C_{\text{obs}})$ at weight w_{obs}, and J ridge rows $(\mathbf{0}, I_J)$ at weight λ_k, against the response $(\tilde{y}^k,\ y^k_{\text{obs}},\ \mathbf{0})$:

$$A = \begin{bmatrix} X & \mathbf{0} \\ X_{\text{obs}} & C_{\text{obs}} \\ \mathbf{0} & I_J \end{bmatrix}, \qquad y = \begin{bmatrix} \tilde{y}^k \\ y^k_{\text{obs}} \\ \mathbf{0} \end{bmatrix}, \qquad W = \text{diag}\big(\tfrac{n_k}{N}\mathbf{1}_N,\ w_{\text{obs}},\ \lambda_k \mathbf{1}_J\big). \tag{5}$$

The first block is the g-prior, with the county columns zeroed so that it affects β only. The second is the reported data, with the share equations weighted by counted ballots — normalized to mean one, so n_k keeps its interpretation in precinct-equivalents — and the turnout equation unweighted. The third shrinks u toward zero with weight λ_k; it is these rows, not C itself, that make the county coefficients a random effect. The whole system is one weighted least squares solve, which keeps conjugacy and closed-form draws.

The covariance matrix. Σ is assembled as $D\,\mathcal{C}\,D$: the nine per-equation residual standard deviations D come from the current night's fits, while the correlations $\mathcal{C}$ are supplied from past elections, because the modes report in sequence and the cross-mode blocks are never estimable from tonight's data. The measured within-mode correlation between turnout and share varies in sign across modes, so a single pooled value will not do.

8 Prediction

Take 500 posterior draws. Transform each draw into the relevant units — $\kappa n_i \cdot \text{logit}^{-1}(\theta)$ for the precinct total, softmax for the mode composition, and logit^{-1} for the two shares — so that every

draw satisfies the adding-up and bound constraints by construction. Then combine counted and remaining votes per cell using (2) and (3), and aggregate.